

CHENYE YANG

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EDUCATION

University of California, Davis - United States

Ph.D. Candidate in Electrical and Computer Engineering

Sep 2022 – Present

Degree GPA: 3.87/4.0

Columbia University - United States

M.S. Advanced Research Specialization in Electrical Engineering

Nikola Tesla Electrical Engineering Scholarship

Sep 2019 – Apr 2021

GPA: 3.8/4.0

Xi'an Jiaotong University - China

B.Eng. in Automation

Honors Electronic and Information Engineering Program (QianXuesen Class)

Sep 2015 – Jun 2019

Grade: 89.42/100

RESEARCH EXPERIENCE

Robust Rubric-based Preference Design for DPO/GRPO

Prof. Lifeng Lai, UC Davis, May 2026 – Present

- Study the inherent robustness of rubric-based preference models for DPO/GRPO.
- Design robust rubric-based preference models to mitigate the impact of adversarial preference poisoning.

Adversarially Robust DPO

Prof. Lifeng Lai, UC Davis, Jan 2026 – Present

- Study adversarial robustness of DPO against preference poisoning from a theoretical perspective.
- Develop gradient-space analysis showing how flipped pairwise comparisons perturb log-linear DPO.
- Propose Trim- K and Soft-Trim- K Robust-DPO, which remove or downweight high-impact preference pairs.
- Prove worst-case gradient-bias, parameter- and policy-deviation guarantees for the proposed robust DPO algorithms.
- Conduct experiments on the SHP dataset to validate the theoretical robustness behavior. (*Manuscript under review*)

Preference Poisoning Attack on DPO

Prof. Lifeng Lai, UC Davis, May 2025 – Dec 2025

- Studied targeted preference poisoning attacks against offline RLHF/DPO from a theoretical perspective.
- Showed that label flipping in log-linear DPO induces a parameter-independent gradient shift, reducing the attack design to a binary sparse approximation problem.
- Proposed two efficient attack algorithms: Binary-Aware Lattice Attack and Binary Matching Pursuit Attack.
- Proved theoretical guarantees for attack feasibility, binary enforcement, minimum-flip recovery, coherence-based support recovery, and impossibility of successful K -flip attacks.
- Experiments on synthetic data and the SHP dataset validate the theory. (*Accepted to ICML 2026*)

Preference Robustness of Online RLHF

Prof. Lifeng Lai, UC Davis, Dec 2023 – May 2025

- Studied preference robustness of RLHF algorithms with online human feedback from a theoretical perspective.
- Investigated preference attacks that force online RLHF to learn a target suboptimal policy with small attack cost.
- Proposed an adversarial human-feedback poisoning strategy and proved its success in misleading online RLHF.
- Proposed a robust-defense online RLHF and proved its robustness to any attacker with bounded attack cost.
- Simulation results validate the theoretical analysis. (*Published in IEEE TSP 2025*)

Reward Robustness of Bandit

Prof. Lifeng Lai, UC Davis, Sep 2022 – Feb 2024

- Studied reward attacks on stochastic multi-armed bandits in non-stationary environments.
- Investigated attackers who force the victim algorithm to mostly choose a suboptimal arm with small attack cost.
- Considered three general scenarios with different environments, algorithms, and attacker information.
- Proposed three attack strategies and proved their success in terms of target-arm selection and attack cost.
- Proposed a non-stationary bandit defense method and proved its robustness to any attacker with bounded attack cost.
- Simulation results validate the theoretical analysis. (*Accepted to ACSSC 2023 and Published in IEEE TSP 2024*)

Edge AI System for Smart Home

Prof. Xiaofan (Fred) Jiang, Columbia University, Jun 2021 – Sep 2021

Summer Research Assistant in Columbia Intelligent and Connected Systems Lab.

- An auto-discover system able to allocate sensors and actuators according to the smart home applications user specified; deployed in multiple lab rooms. (*Accepted to SenSys 2021*)
- Designed the MCU/sensors-to-application architecture and built the hardware testbed from scratch.

Fever Screening System

Prof. Xiaofan (Fred) Jiang, Columbia University, Jan 2020 – May 2021

Research Intern in Columbia Intelligent and Connected Systems Lab.

- A low-cost system based on RGB-thermal cameras for continuous fever screening of multiple people without human interaction; deployed in a restaurant and hospital clinic in New York City. Featured in Columbia News.
- Implemented multiple RGB-thermal heads matching, trained and deployed YOLOV3 (head detection) and FSA-Net (head orientation regression), estimated distance using non-identical RGB and thermal camera.
- (Accepted to SenSys 2020, CPHS 2021, and IPSN 2022)

Khameleon Scheduler in RL

Prof. Eugene Wu, Columbia University, Jul 2020 – Sep 2020

Research Intern in WuLab Columbia University.

- A server-side scheduler involving a complex optimization based on available resources, predicted user interactions, and response quality levels to maximize user-perceived interactivity and satisfaction in real-time.
- Implemented Q-learning and SARSA-based prefetching schedulers in a simulated RL environment to trade off latency and response quality in cloud-based interactive applications.

PUBLICATIONS

- [1] **Chenye Yang**, Weiyu Xu, Lifeng Lai, “Adversarially Robust Direct Preference Optimization against Preference Label Flip Attacks” *Manuscript under review*.
- [2] Mo Lyu, **Chenye Yang**, Lifeng Lai, “Efficient Reward Manipulation Attacks Against GRPO under RLVR” *Manuscript under review*.
- [3] **Chenye Yang**, Weiyu Xu, Lifeng Lai, “Efficient Preference Poisoning Attack on Offline RLHF” *The 43rd International Conference on Machine Learning (ICML 2026)*.
- [4] Mo Lyu, **Chenye Yang**, Guanlin Liu, Lifeng Lai, “Dueling Bandit: Adversarial Attack and Robust Defense” *Manuscript under review*.
- [5] Mo Lyu, **Chenye Yang**, Guanlin Liu, Lifeng Lai, “Adversarial Post-Action Attacks on Dueling Bandits” *The 59th Asilomar Conference on Signals, Systems, and Computers (ACSSC 2025)*.
- [6] **Chenye Yang**, Mo Lyu, Guanlin Liu, Lifeng Lai, “Human Feedback Attack on Online RLHF: Attack and Robust Defense” *IEEE Transactions on Signal Processing, Volume: 73 (IEEE TSP 2025)*.
- [7] **Chenye Yang**, Guanlin Liu, Lifeng Lai, “Stochastic Bandits With Non-Stationary Rewards: Reward Attack and Defense” *IEEE Transactions on Signal Processing, Volume: 72 (IEEE TSP 2024)*.
- [8] **Chenye Yang**, Guanlin Liu, Lifeng Lai, “Reward Attack on Stochastic Bandits with Non-stationary Rewards” *The 57th Asilomar Conference on Signals, Systems, and Computers (ACSSC 2023)*.
- [9] Kaiyuan Hou, Yanchen Liu, Peter Wei, **Chenye Yang**, Hengjiu Kang, Stephen Xia, Teresa Spada, Andrew Rundle, Xiaofan Jiang, “A Low-Cost In-situ System for Continuous Multi-Person Fever Screening” *The 21st ACM/IEEE International Conference on Information Processing in Sensor Networks (IPSN 2022)*.
- [10] Stephen Xia, Rishikanth Chandrasekaran, Yanchen Liu, **Chenye Yang**, Tajana Simunic Rosing, Xiaofan Jiang, “Demo Abstract: A Drone-based System for Intelligent and Autonomous Homes” *The 19th ACM Conference on Embedded Networked Sensor Systems (SenSys 2021), Best Demo Award*.
- [11] Peter Wei, Yanchen Liu, Hengjiu Kang, **Chenye Yang**, Xiaofan Jiang, “A Low-Cost and Scalable Personalized Thermal Comfort Estimation System in Indoor Environments” *The 1st International Workshop on Cyber-Physical-Human System Design and Implementation (CPHS 2021)*.
- [12] Peter Wei, **Chenye Yang**, Xiaofan Jiang, “Poster Abstract: Low-Cost Multi-Person Continuous Skin Temperature Sensing System for Fever Detection” *The 18th ACM Conference on Embedded Networked Sensor Systems (SenSys 2020)*.
- [13] Pengyuan Liu, **Chenye Yang**, Jiang Wu, “Hybrid Features Based K-means Clustering Algorithm for Use in Electricity Customer Load Pattern Analysis” *The 37th Chinese Control Conference (CCC 2018)*.

PROJECT EXPERIENCE

Error Correcting Code

Prof. Ulrich Jetzek (from Fachhochschule Kiel), UC Davis, Apr 2023 – Jun 2023

- Converted TXT/PNG/WAV files to bit streams; encoded with (7,4) Hamming and (n, k) systematic cyclic codes.
- Simulated binary symmetric channels; decoded with syndrome/error-trapping decoders and reconstructed files.

Large Scale Stream Processing

Prof. Deepak S. Turaga, Columbia University, Jan 2020 – May 2020

- Built a multithreaded Python data generator and stream processor with a database-backed web control interface.
- Designed socket/HTTP data paths and WebSocket control channels across components.

Sparse Models for High-Dimensional Data

Prof. John Wright, Columbia University, Jan 2020 – May 2020

- Built a system for fast moving-object detection in high-resolution video; tested on sports video and daily scenes.
- Used super-resolution reconstruction to speed processing while preserving high-resolution output.

Internet of Things

Prof. Xiaofan (Fred) Jiang, Columbia University, Sep 2019 – Dec 2019

- Programmed ESP8266 for I/O, APIs, server mode, MongoDB connectivity, and gesture recognition application.
- Built a distributed scalable system to monitor soil conditions over long time at multiple locations via LoRa wireless communication; data stored in MongoDB; website designed for data visualization and systems control.

Statistical Learning

Prof. Predrag Jelenkovic, Columbia University, Sep 2019 – Dec 2019

- Reproduced the results of identification/authentication in *Smartphone and Smartwatch-Based Biometrics Using Activities of Daily Living* with WISDM Dataset using decision tree, random forest and K-NN.
- Trained SVM on the same dataset and compare different methods.

Random Matrix Theory and Application

Prof. Ori Shental, Columbia University, Sep 2019 – Dec 2019

- Reproduced the results of uniform samples in *Bandlimited Field Reconstruction for Wireless Sensor Networks*.
- Tested Gaussian sampling and analyzed the resulting differences.

TEACHING EXPERIENCE

Ph.D.	Reader	EEC 161 Applied Probability	Winter 2026
	Teaching Assistant	EEC 256 Reinforcement Learning	Spring 2025
		ENG 006 Engineering Problem Solving	Winter 2025
		EEC 173A / ECS 152A Computer Networks	Winter 2024 & Winter 2023
M.S.	Teaching Assistant	E4764 IoT Intelligent and Connected Systems	Fall 2020

WORK EXPERIENCE

WiLO Networks Inc. Network Engineer Intern

Oct 2021 – Apr 2022

Early-stage startup; owned end-to-end development of the smart-farm prototype across hardware, network, and cloud.

- Architected and implemented a low-cost LoRa mesh network on ESP32 + SX1276 (LoRa PHY), including auto-join/routing protocol and robustness strategies; validated via Duke Smart Farm field tests and NS-3 simulation.
- Integrated and modified MCCI LoRaWAN LMIC at the register level to control SX1276 mode transitions, interrupts, and Tx/Rx scheduling; aligned with ESP32 OS.Job timing for reliable communications.
- Delivered sensor-to-cloud data pipeline (sensors → MCU → LoRa → Raspberry Pi gateway → InfluxDB/Azure) enabling data fusion and ML-based monitoring.
- Standardized reusable MCU libraries + KiCad schematic/footprint libraries for team-wide reuse, and designed and assembled a backscatter-tag PCB in KiCad (component selection + layout).

TECHNICAL SKILLS

Programming:	Python, C/C++/SystemC, MATLAB, HTML, R, Verilog HDL
Software & Tools:	AI Coding: OpenAI Codex, Claude Code
	Platform: VS Code, PlatformIO
	Design: KiCad, Autodesk Inventor, Altium Designer, LabVIEW, FDTD Solutions
	Others: GitHub, Linux, LaTeX, Docker, PyTorch, OpenCV, Spark, AWS, GCP

AWARDS

Gold Reviewer Award	<i>The 43rd International Conference on Machine Learning (ICML)</i>	2026
Best Demo Award	<i>The 19th ACM Conference on Embedded Networked Sensor Systems (SenSys)</i>	2021
Nikola Tesla Scholarship	<i>Columbia University</i>	2019
University Third Scholarship	<i>Xi'an Jiaotong University</i>	2018
Siyuan Scholarship	<i>Xi'an Jiaotong University</i>	2017
Outstanding Student	<i>Xi'an Jiaotong University</i>	2017
Provincial First Prize	<i>China Undergraduate Mathematical Contest in Modeling</i>	2016
National Second Prize	<i>VEX Robotics China Open</i>	2016